

21) Which rules of inference are used to establish the conclusion of Lewis Carroll's argument described in Example 26 of Section 1.4?

Soln: Premise: (1) $\forall x(P(x) \rightarrow Q(x))$ (2) $\exists x(P(x) \wedge \neg R(x))$

Conclusion: $\exists x(Q(x) \wedge \neg R(x))$

Steps

1. $\forall x(P(x) \rightarrow Q(x))$
2. $\exists x(P(x) \wedge \neg R(x))$
3. $P(c) \wedge \neg R(c)$ for some c in the domain
4. $P(c)$
5. $P(c) \rightarrow Q(c)$
6. $Q(c)$
7. $\neg R(c) \wedge P(c)$
8. $\neg R(c)$
9. $Q(c) \wedge \neg R(c)$
10. $\exists x(Q(x) \wedge \neg R(x))$

Reason

Premise

Premise

Existential instantiation of 2

Simplification on 3

Universal instantiation of 2

Modus ponens on 4, 5

Commutative law on 3

Simplification on 7

Conjunction on 6, 8

Existential generalization on 9.

22) Which rules of inference are used to establish the conclusion of Lewis Carroll's argument described in Example 27 of Section 1.4?

Soln: Premise: 1. $\forall x(P(x) \rightarrow S(x))$ 2. $\neg \exists x(Q(x) \wedge R(x))$ 3. $\forall x(\neg R(x) \rightarrow \neg S(x))$

Conclusion: $\forall x(P(x) \rightarrow \neg Q(x))$

Steps

1. $\forall x(P(x) \rightarrow S(x))$
2. $\neg \exists x(Q(x) \wedge R(x))$
3. $\forall x(\neg R(x) \rightarrow \neg S(x))$

Reason

Premise

Premise

Premise